

# Fonts and Alignment Filter Demonstration

Nandakumar Chandrasekhar

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This document demonstrates every feature provided by the `fonts-and-alignment` Lua filter. The syntax shown in the code blocks directly produces the output below them, and the same Markdown source produces faithful output in both LaTeX/PDF and HTML.

## 1 Font Sizing Scale (Inline)

Nine sizing hooks scale relative to the document's base font size.

| Size         | Syntax                                  | Output |
|--------------|-----------------------------------------|--------|
| tiny         | <code>[Sample]{.pfa-text-3xs}</code>    | Sample |
| scriptsize   | <code>[Sample]{.pfa-text-2xs}</code>    | Sample |
| footnotesize | <code>[Sample]{.pfa-text-xs}</code>     | Sample |
| small        | <code>[Sample]{.pfa-text-s}</code>      | Sample |
| normal       | <code>[Sample]{.pfa-text-normal}</code> | Sample |
| large        | <code>[Sample]{.pfa-text-l}</code>      | Sample |
| Large        | <code>[Sample]{.pfa-text-xl}</code>     | Sample |
| LARGE        | <code>[Sample]{.pfa-text-2xl}</code>    | Sample |
| huge         | <code>[Sample]{.pfa-text-3xl}</code>    | Sample |

## 2 Font Sizing Scale (Block)

The same sizing classes apply to fenced div blocks. This is useful for sidebars, captions, or callouts.

```
::: pfa-text-s
```

This entire paragraph renders at the `small` size. Block-level sizing

↪ ensures the content consistently shrinks or grows as a unified

↪ component.

```
:::
```

This entire paragraph renders at the `small` size. Block-level sizing ensures the content consistently shrinks or grows as a unified component.

```
::: pfa-text-l
```

This entire paragraph renders at the ``large`` size — well suited for pull  
↪ quotes or emphasized opening text.

```
:::
```

This entire paragraph renders at the `large` size — well suited for pull quotes or emphasized opening text.

### 3 Font Weights, Shapes, and Families (Inline)

Every supported typographic style is available as an inline span constraint.

| Style        | Syntax                                     | Output        |
|--------------|--------------------------------------------|---------------|
| Bold         | <code>[Sample]{.pfa-font-bold}</code>      | <b>Sample</b> |
| Medium       | <code>[Sample]{.pfa-font-medium}</code>    | Sample        |
| Italic       | <code>[Sample]{.pfa-font-italic}</code>    | <i>Sample</i> |
| Slanted      | <code>[Sample]{.pfa-font-slanted}</code>   | <i>Sample</i> |
| Upright      | <code>[Sample]{.pfa-font-upright}</code>   | Sample        |
| Emphasis     | <code>[Sample]{.pfa-font-emphasis}</code>  | <i>Sample</i> |
| Serif        | <code>[Sample]{.pfa-font-serif}</code>     | Sample        |
| Sans-Serif   | <code>[Sample]{.pfa-font-sans}</code>      | Sample        |
| Monospace    | <code>[Sample]{.pfa-font-mono}</code>      | Sample        |
| Small Caps   | <code>[Sample]{.pfa-font-smallcaps}</code> | SAMPLE        |
| Normal Reset | <code>[Sample]{.pfa-font-normal}</code>    | Sample        |

### 4 Font Weights, Shapes, and Families (Block)

The same classes work at the block level, applying the typographic voice across all internal sentences without requiring inline markers.

```
::: pfa-font-sans
```

This paragraph is rendered using the document's sans-serif family.

↪ Block-scoped family switches are convenient for code blocks or  
↪ structural sections.

```
:::
```

This paragraph is rendered using the document's sans-serif family. Block-scoped family switches are convenient for code blocks or structural sections.

```
::: pfa-font-smallcaps
```

Section headings or short callouts can be rendered entirely in small caps

↪ using the block form.

```
:::
```

SECTION HEADINGS OR SHORT CALLOUTS CAN BE RENDERED ENTIRELY IN SMALL CAPS  
USING THE BLOCK FORM.

## 5 Text Casing Transformations

Casing classes rewrite the underlying Abstract Syntax Tree (AST) text nodes rather than applying CSS formatting, meaning the transformation survives raw copy-paste operations.

### 5.1 Inline Casing

| Casing    | Syntax                    | Output  |
|-----------|---------------------------|---------|
| Uppercase | [coerced]{.pfa-uppercase} | COERCED |
| Lowercase | [COERCED]{.pfa-lowercase} | coerced |

### 5.2 Block Casing

```
::: pfa-uppercase
```

this block is uniformly uppercased, even across paragraph boundaries.

```
:::
```

THIS BLOCK IS UNIFORMLY UPPERCASED, EVEN ACROSS PARAGRAPH BOUNDARIES.

## 6 Inline Text Decorations

Underline and strikethrough variants are routed through the `ulem` LaTeX package and mapped to matching standard CSS.

| Decoration       | Syntax                                            | Output               |
|------------------|---------------------------------------------------|----------------------|
| Single Underline | <code>[Sample]{.pfa-text-underline}</code>        | <u>Sample</u>        |
| Double Underline | <code>[Sample]{.pfa-text-underline-double}</code> | <u><u>Sample</u></u> |
| Dashed Underline | <code>[Sample]{.pfa-text-underline-dashed}</code> | <u>Sample</u>        |
| Dotted Underline | <code>[Sample]{.pfa-text-underline-dotted}</code> | <u>Sample</u>        |
| Wavy Underline   | <code>[Sample]{.pfa-text-underline-wave}</code>   | <u>Sample</u>        |
| Strikethrough    | <code>[Sample]{.pfa-text-strikethrough}</code>    | <del>Sample</del>    |
| Marked Out       | <code>[Sample]{.pfa-text-markout}</code>          | <del>Sample</del>    |

## 7 Color Resolution

The `pfa-font-color` attribute natively parses standard CSS3 named colors, six-digit hex codes, and three-digit hex shorthands.

| Input Type    | Syntax                                          | Output |
|---------------|-------------------------------------------------|--------|
| CSS Named     | <code>[Sample]{pfa-font-color="crimson"}</code> | Sample |
| Hex Full      | <code>[Sample]{pfa-font-color="#2E8B57"}</code> | Sample |
| Hex Shorthand | <code>[Sample]{pfa-font-color="#666"}</code>    | Sample |

### 7.1 Flexible Spelling of Named Colors

The engine sanitizes inputs automatically. It is case-insensitive and ignores spaces, hyphens, and underscores. All syntaxes below successfully map to the standard `mediumvioletred`.

| Variant    | Syntax                                                   | Output |
|------------|----------------------------------------------------------|--------|
| Lowercase  | <code>[Color]{pfa-font-color="mediumvioletred"}</code>   | Color  |
| Title Case | <code>[Color]{pfa-font-color="MediumVioletRed"}</code>   | Color  |
| Spaces     | <code>[Color]{pfa-font-color="Medium Violet Red"}</code> | Color  |

| Variant | Syntax                                                   | Output |
|---------|----------------------------------------------------------|--------|
| Hyphens | <code>[Color]{pfa-font-color="medium-violet-red"}</code> | Color  |

## 7.2 Block-Level Color Constraints with Inline Overrides

Colors applied to a block automatically inherit downward, but nested spans can explicitly override the parent value.

```
::: {pfa-font-color="#666"}
```

The parent block carries a default color of dark slate gray. `[This inline span overrides the color to tomato red.]``{pfa-font-color="tomato"}` The surrounding text continues in the parent color until the block closes.

```
:::
```

The parent block carries a default color of dark slate gray. **This inline span overrides the color to tomato red.** The surrounding text continues in the parent color until the block closes.

## 8 Text Alignment within Blocks

The `.pfa-text-*` utility wraps blocks in standard directional text environments. The blocks below are intentionally long so that the wrapped lines reveal the edge behavior of each alignment.

```
::: pfa-text-left
```

This block is left-aligned. Words begin flush against the left margin and the right edge falls wherever the natural word boundaries land, producing the relaxed silhouette typically used for body prose in left-to-right scripts.

```
:::
```

This block is left-aligned. Words begin flush against the left margin and the right edge falls wherever the natural word boundaries land, producing the relaxed silhouette typically used for body prose in left-to-right scripts.

```
::: pfa-text-center
```

This block is centered horizontally within the text column. Centered prose is best reserved for short callouts, pull quotes, dedications, or titlecards — sustained reading at this alignment is difficult because the eye cannot anchor on a consistent margin.

```
:::
```

This block is centered horizontally within the text column. Centered prose is best reserved for short callouts, pull quotes, dedications, or titlecards — sustained reading at this alignment is difficult because the eye cannot anchor on a consistent margin.

```
::: pfa-text-right
```

This block is right-aligned. The right edge stays flush against the column boundary while words spill back toward the left, an effect suited to sidenotes, attributions, and short captions placed against a column edge.

```
:::
```

This block is right-aligned. The right edge stays flush against the column boundary while words spill back toward the left, an effect suited to sidenotes, attributions, and short captions placed against a column edge.

## 9 Ragged Alignment (Line-Break Honoring)

The `.pfa-align-*` utility preserves and honors explicit line breaks (via backslash escapes), making it ideal for poetry, formal addresses, and titles. In PDF, the two families also differ in vertical spacing: `.pfa-text-*` mirrors LaTeX's `center/flushleft/flushright` environments and introduces vertical separation around the block, while `.pfa-align-*` mirrors the `\centering/\raggedright/\raggedleft` declarations and flows inline with the surrounding content. HTML applies only `text-align` to both families, so the vertical-spacing distinction is PDF-only.

Choose `.pfa-text-*` when the aligned block is its own visual unit (a callout, a centered pull quote, a standalone announcement) and `.pfa-align-*` when the alignment is part of the surrounding paragraph flow (a line-broken signature, a verse stanza, a captioned figure).

```
::: pfa-align-left
First left-aligned line\
Second left-aligned line\
Third left-aligned line
:::
```

First left-aligned line  
Second left-aligned line  
Third left-aligned line

```
::: pfa-align-center
First centered line\
Second centered line\
Third centered line
:::
```

First centered line  
Second centered line  
Third centered line

```
::: pfa-align-right
First right-aligned line\
Second right-aligned line\
Third right-aligned line
:::
```

First right-aligned line  
Second right-aligned line  
Third right-aligned line

## 10 Block-Level Alignment

The `.pfa-block-*` utility wraps content in an isolated bounding box and anchors the box to the left margin, the center of the text column, or the right margin. The box width is set by the widest single line inside the block, so explicit line breaks (`\`) shape both the silhouette and the horizontal footprint of the panel.

```
::: {.pfa-block-left}
```

A self-contained advisory hugging the left margin of the column.\  
The bounding box is set by the widest line of internal content.\  
Surrounding paragraphs continue without being pushed sideways.

```
:::
```

A self-contained advisory hugging the left margin of the column.  
The bounding box is set by the widest line of internal content.  
Surrounding paragraphs continue without being pushed sideways.

```
::: {.pfa-block-center}
```

Useful for callouts, banners, or section dividers that should\  
catch the reader's eye without committing to the full column.\  
The panel centers itself relative to the surrounding text.

```
:::
```

Useful for callouts, banners, or section dividers that should  
catch the reader's eye without committing to the full column.  
The panel centers itself relative to the surrounding text.

```
::: {.pfa-block-right}
```

Anchored against the right margin like a marginal sidenote.\  
Ideal for attributions, version stamps, or supplementary notes\  
that should sit beside the main text rather than within it.

```
:::
```

Anchored against the right margin like a marginal sidenote.  
Ideal for attributions, version stamps, or supplementary notes  
that should sit beside the main text rather than within it.

## 11 Combining Multiple Classes

Classes compose seamlessly. Sizing, family, weight, color, and decoration definitions can stack safely on any single element.

### 11.1 Inline Composition



| Style               | Syntax                                                                                      | Output        |
|---------------------|---------------------------------------------------------------------------------------------|---------------|
| Bold Sans Display   | <code>[Sample]{.pfa-font-bold .pfa-font-sans .pfa-text-l pfa-font-color="red"}</code>       | <b>Sample</b> |
| Italic Mono Caption | <code>[Sample]{.pfa-font-italic .pfa-font-mono .pfa-text-s}</code>                          | <i>Sample</i> |
| Smallcaps Underline | <code>[Sample]{.pfa-text-underline .pfa-font-smallcaps pfa-font-color="forestgreen"}</code> | <u>SAMPLE</u> |

## 11.2 Block Composition

```
::: {.pfa-text-center .pfa-font-sans .pfa-font-bold .pfa-text-l
↪ pfa-font-color="midnightblue"}
```

A centered, bold, sans-serif, large, midnight-blue block assembled from  
↪ five utilities.

```
:::
```

**A centered, bold, sans-serif, large, midnight-blue block assembled from five utilities.**

## 12 Legacy Short-Hand Aliases (Deprecated)

**Deprecated** — retained strictly for backward compatibility and will be removed in the next major release. New documents should use the explicit `pfa-*` namespaces above to prevent global framework collisions.

### 12.1 Font Weight, Shape, and Family Aliases

| Legacy Alias            | Short           | Modern Class                   |
|-------------------------|-----------------|--------------------------------|
| <code>bold</code>       | <code>bf</code> | <code>pfa-font-bold</code>     |
| <code>emphasis</code>   | <code>em</code> | <code>pfa-font-emphasis</code> |
| <code>italic</code>     | <code>it</code> | <code>pfa-font-italic</code>   |
| <code>medium</code>     | <code>md</code> | <code>pfa-font-medium</code>   |
| <code>monospace</code>  | <code>tt</code> | <code>pfa-font-mono</code>     |
| <code>normalfont</code> | <code>nf</code> | <code>pfa-font-normal</code>   |
| <code>sans</code>       | <code>sf</code> | <code>pfa-font-sans</code>     |

| Legacy Alias | Short | Modern Class       |
|--------------|-------|--------------------|
| serif        | rm    | pfa-font-serif     |
| slanted      | sl    | pfa-font-slanted   |
| smallcaps    | sc    | pfa-font-smallcaps |
| upright      | up    | pfa-font-upright   |

## 12.2 Font Size Aliases

| Legacy Alias | Modern Class    |
|--------------|-----------------|
| xsmall       | pfa-text-xs     |
| small        | pfa-text-s      |
| normal       | pfa-text-normal |
| large        | pfa-text-l      |
| xlarge       | pfa-text-xl     |
| xxlarge      | pfa-text-2xl    |
| huge         | pfa-text-3xl    |

## 12.3 Alignment Aliases

| Legacy Alias | Modern Class     |
|--------------|------------------|
| center       | pfa-text-center  |
| flushleft    | pfa-text-left    |
| flushright   | pfa-text-right   |
| centering    | pfa-align-center |
| raggedleft   | pfa-align-right  |
| raggedright  | pfa-align-left   |

## 12.4 Underline and Strikeout Aliases

| Legacy Alias | Short | Modern Class              |
|--------------|-------|---------------------------|
| uline        | u     | pfa-text-underline        |
| uuline       | uu    | pfa-text-underline-double |
| dashuline    | dau   | pfa-text-underline-dashed |
| dotuline     | dou   | pfa-text-underline-dotted |
| uwave        | uw    | pfa-text-underline-wave   |
| sout         | so    | pfa-text-strikeout        |

| Legacy Alias | Short | Modern Class     |
|--------------|-------|------------------|
| xout         | xo    | pfa-text-markout |